SIMATS Engineering • CSE • CSA0802 — Python Programming • 05-08-2026 • Category: Class Work (4.8.26) • Student Submissions Report

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Q1. Union of Sets

Score: 0.00 **TC: 0/1 passed**

CODE

```
A = set(map(int, input().split()))
B = set(map(int, input().split()))
print(A.union(B))
```

INPUT

```
1 2 3
3 4 5
```

OUTPUT

```
{1, 2, 3, 4, 5}
```

Q2. Intersection

Score: 0.00 **TC: 0/1 passed**

CODE

```
A = set(map(int, input().split()))
B = set(map(int, input().split()))
print(A.intersection(B))
```

INPUT

```
1 2 3
2 3 4
```

OUTPUT

```
{2, 3}
```

Q3. Difference

Score: 0.00 **TC: 0/1 passed**

CODE

```
A = set(map(int, input().split()))
B = set(map(int, input().split()))
print(A - B)
```

INPUT

```
1 2 3
2 3 4
```

OUTPUT

```
{1}
```

Q4. Symmetric Difference

Score: 0.00 **TC: 0/1 passed**

CODE

```
A = set(map(int, input().split()))
B = set(map(int, input().split()))
print(A.symmetric_difference(B))
```

INPUT

```
1 2 3
2 3 4
```

OUTPUT

```
{1, 4}
```

Q5. Subset

Score: 0.00 **TC: 0/1 passed**

CODE

```
A = set(map(int, input().split()))
B = set(map(int, input().split()))
if A.issubset(B):
    print("Subset")
else:
    print("Not Subset")
```

INPUT

```
2 3
1 2 3 4
```

OUTPUT

```
Subset
```

Q6. Superset

Score: 0.00 **TC: 0/1 passed**

CODE

```
A = set(map(int, input().split()))
B = set(map(int, input().split()))
if A.issuperset(B):
    print("Superset")
else:
    print("Not Superset")
```

INPUT

```
1 2 3 4
2 3
```

OUTPUT

```
Superset
```

Q7. Disjoint Sets

Score: 0.00 **TC: 0/1 passed**

CODE

```
a = set(map(int, input().split()))
b = set(map(int, input().split()))
if a.isdisjoint(b):
    print("Disjoint")
else:
    print("Not Disjoint")
```

INPUT

```
1 2
3 4
```

OUTPUT

```
Disjoint
```

Q8. Remove Duplicates

Score: 0.00 **TC: 0/1 passed**

CODE

```
lst = list(map(int, input().split()))
print(tuple(set(lst)))
```

INPUT

```
1 2 2 3 3 4
```

OUTPUT

```
(1, 2, 3, 4)
```

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Q9. Common Elements

Score: 0.00 **TC: 0/1 passed**

CODE

```
a = set(map(int, input().strip("(){}").split(",")))
b = set(map(int, input().strip("(){}").split(",")))
c = set(map(int, input().strip("(){}").split(",")))
print(a & b & c)
```

INPUT

```
(1,2,3)
(2,3,4)
(2,5)
```

OUTPUT

```
{2}
```

Q10. Missing Numbers

Score: 0.00 **TC: 0/1 passed**

CODE

```
given = {1, 2, 4, 5, 7, 10}
all_numbers = set(range(1, 11))
print(all_numbers - given)
```

INPUT

```
{3, 6, 8, 9}
```

OUTPUT

```
{8, 9, 3, 6}
```

Q11. First Repeated

Score: 0.00 **TC: 0/1 passed**

CODE

```
lst = [10, 20, 30, 20, 40]
seen = set()
for num in lst:
    if num in seen:
        print("First repeated element is:", num)
        break
    seen.add(num)
else:
    print("No repeated element found")
```

INPUT

```
10,20,30,20,40
```

OUTPUT

```
First repeated element is: 20
```

Q12. Duplicate Elements

Score: 0.00 **TC: 0/1 passed**

CODE

```
a = list(map(int, input().split()))
seen = set()
dup = []
for i in a:
    if i in seen and i not in dup:
        dup.append(i)
    else:
        seen.add(i)
print(tuple(dup))
```

INPUT

```
10 20 30 20 10
```

OUTPUT

```
(20, 10)
```

Q13. Compare Lists

Score: 0.00 **TC: 0/1 passed**

CODE

```
list1 = list(map(int, input().split()))
list2 = list(map(int, input().split()))
if set(list1) == set(list2):
    print(True)
else:
    print(False)
```

INPUT

```
1 2 2 3
3 2 1
```

OUTPUT

```
True
```

Q14. Duplicate Tuples

CODE

```
l = eval(input())
s = set()
res = []
for i in l:
    if i not in s:
        s.add(i)
        res.append(i)
print(res)
```

INPUT

```
[(1,2),(1,2),(3,4)]
```

OUTPUT

```
(no output)
```

Q15. Pair Sum

Score: 0.00 **TC: 0/1 passed**

CODE

```
lst = list(map(int, input().split()))
target = int(input())
for i in range(len(lst)):
    for j in range(i + 1, len(lst)):
        if lst[i] + lst[j] == target:
            print((lst[i], lst[j]), end="")
```

INPUT

```
2 4 6 8 10
12
```

OUTPUT

```
(2, 10)(4, 8)
```

Q16. Unique Characters

Score: 5.00 **TC: 1/1 passed**

CODE

```
s = input().strip()
print(len(set(s)))
```

INPUT

```
Programming
```

OUTPUT

```
8
```

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Q17. Unique Words

Score: 0.00 **TC: 0/1 passed**

CODE

```
sentence = input()
words = sentence.split()
unique_words = set(words)
print(unique_words)
```

INPUT

```
python is easy python
```

OUTPUT

```
{'is', 'easy', 'python'}
```

Q18. Exactly One Set

Score: 0.00 **TC: 0/1 passed**

CODE

```
a = set(map(int, input().split()))
b = set(map(int, input().split()))
c = set(map(int, input().split()))
result = (a - b - c) | (b - a - c) | (c - a - b)
print(result)
```

INPUT

```
1 2 3
3 4 5
5 6 7
```

OUTPUT

```
{1, 2, 4, 6, 7}
```

Q19. Menu Driven

CODE

```
s1 = set(map(int, input().split()))
s2 = set(map(int, input().split()))
while True:
    print("1. Union")
    print("2. Intersection")
    print("3. Difference (A-B)")
    print("4. Symmetric Difference")
    print("5. Exit")
    ch = int(input())
    if ch == 1:
        print("Union:", s1 | s2)
    elif ch == 2:
        print("Intersection:", s1 & s2)
    elif ch == 3:
        print("Difference:", s1 - s2)
    elif ch == 4:
        print("Symmetric Difference:", s1 ^ s2)
    elif ch == 5:
        print("Program terminated.")
        break
    else:
        print("Invalid choice")
```

INPUT

```
1 2 3 4
3 4 5 6
1
2
3
4
5
```

OUTPUT

```
1. Union
2. Intersection
3. Difference (A-B)
4. Symmetric Difference
5. Exit
Union: {1, 2, 3, 4, 5, 6}
1. Union
2. Intersection
3. Difference (A-B)
4. Symmetric Difference
5. Exit
Intersection: {3, 4}
1. Union
2. Intersection
3. Difference (A-B)
4. Symmetric Difference
5. Exit
Difference: {1, 2}
1. Union
2. Intersection
3. Difference (A-B)
4. Symmetric Difference
5. Exit
Symmetric Difference: {1, 2, 5, 6}
1. Union
2. Intersection
3. Difference (A-B)
4. Symmetric Difference
5. Exit
Program terminated.
```

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Q20. Student Database

CODE

```
students = set()
while True:
    ch = int(input())
    if ch == 1:
        students.add(input())
        print("Student added successfully.")
    elif ch == 2:
        students.discard(input())
        print("Student removed successfully.")
    elif ch == 3:
        print("Student found." if input() in students else "Student not found.")
    elif ch == 4:
        print("Students:", students)
    elif ch == 5:
        print("Total number of students:", len(students))
    elif ch == 6:
        print("Program terminated.")
        break
```

INPUT

```
1
Ravi
1
Priya
4
3
Ravi
2
Ravi
5
6
```

OUTPUT

```
Student added successfully.
Student added successfully.
Students: {'Ravi', 'Priya'}
Student found.
Student removed successfully.
Total number of students: 1
Program terminated.
```